

ASSESSMENT 2 BRIEF	
Subject Code and Title	DLE602 Deep Learning
Assessment	Deep Learning Project Proposal Presentation
Individual/Group	Group
Length	1,000 words (+/-10%) Report and Presentation
Learning Outcomes	The Subject Learning Outcomes demonstrated by the successful completion of the task below include: b) Compare and select ways to pre-process signals, images, and texts for natural language, speech recognition, and computer vision applications. c) Develop critical analysis skills in deep learning research. d) Demonstrate collaborative skills to apply deep learning to solve real world problems. e) Demonstrate ability to effectively communicate scientific and technical information.
Submission	Due by 11:55pm AEST/AEDT Sunday end of Module 8.
Weighting	30%
Total Marks	100 marks

Task Summary

In Assessment 1, you read the article entitled 'Deep Convolution Neural Networks for Twitter Sentiment Analysis' by Zhao, Gui and Zhang (2018), which can be accessed at:
<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8244338>.

For Assessment 2, in a small group of two to three people, you are required to select and analyse quality research articles similar to the above-mentioned article and identify an interesting project in the world of Deep Learning. Once you have identified the project, you will present your findings in a professional project proposal report and deliver a presentation.

Please refer to the Task Instructions (below) for further details on how to complete this task.

Context

The aim of this assessment is to develop a project proposal for a hypothetical research project. This report will enable you to demonstrate your ability to search for relevant literature, write about other works in the area and design a research project proposal, which you will implement in Assessment 3.



Finally, you will communicate details of your project (written and oral) to an audience familiar with Deep Learning.

Task Instructions

In a small group of two to three people, you will research a Deep Learning project of your choice and articulate your findings in a written report and a 5–7 minute audio-visual seminar. You can form a group yourself or your learning facilitator will allocate you to a group and advise you of the students with whom you will be working. Your group will comprise students who have selected a similar topic.

You are required to select, assess and evaluate the information that you include in your written project proposal. The presentation will be an audio-visual submission (please note, online students are only required to film a voice over of the presentation and are not required to film themselves delivering the presentation). The 5–7-minute presentation should provide an engaging explanation of your nominated topic and include the most relevant and absorbing information from your written project proposal.

Report Guidelines

In relation to the report, you need to:

- Prepare a 1,000-word report in which you clearly explain the project that you identified in your literature review and demonstrate a sound application of the project management process;
- Provide a concise description of your project in a statement in your report and state the specific aim(s) or research question(s) that you intend to address;
- Present your report using any standard writing format;
- Include a cover page and table of contents. Note: the cover page needs to include official student names and project name;
- Include an abstract/executive summary;
- State the word count at the end of the report (before the reference section);
- Insert page numbers (the page numbers should appear in the footer of each page); and
- Label any figures and tables in your report with meaningful captions.

Presentation Guidelines

In relation to the presentation, you need to:

- Prepare a 5–7 minute audio-visual presentation that summarises your research. All students need to present for an equal amount time;
- Include a title slide with student details/subject details that clearly introduces your group and the research topic;
- Ensure your presentation is engaging and informative. Your presentation should **NOT** include the full details contained in your report;
- Use a PDF format for the presentation slides;



- Keep the layout, text sizes and style consistent throughout the presentation. The presentation should reflect your group dynamic;
- Practice the presentation beforehand to ensure it is clear and concise; and
- Include correct reference lists in the appropriate style.

Referencing

It is essential that you use APA style to cite and reference your research. For more information on referencing, visit our Library website at: <https://library.torrens.edu.au/academicskills/apa/tool>.

Submission Instructions

Compile your written report into a PDF or Word-format and your presentation slides into a PDF format. Submit both the documents using the Assessment link in the main navigation menu for 'DLE602: Deep Learning Assessment 2'.

Face-to-Face Students

You will present a 5–7 minute audio-visual seminar to the class and submit your PDF document via the Assessment link in the main navigation menu for 'DLE602: Deep Learning Assessment 2'. Ensure that you have the names and student ID of the members of the group on the front page. The Learning Facilitator will provide feedback via the Grade Centre in the LMS portal. Feedback can be viewed in My Grades.

Online Students

You will record your presentation using the free Zoom program (available as a free online download). Refer to the 'How to Use Zoom' PDF document located on Blackboard for more information. Ensure that you have the names and student ID of the members of the group on the front page. Record the group presentation and save it, using the following naming convention:

SubjectCodeGroupnumberAssessmentTitle.mp4.

Submit your audio-visual recording and presentation via the Assessment link in the main navigation menu for 'DLE602: Deep Learning Assessment 2'. The Learning Facilitator will provide feedback via the Grade Centre in the LMS portal. Feedback can be viewed in My Grades.

One submission per group is allowed. Use the following naming conventions:

- **SubjectCodeGroupnameAssessmentTitle.pdf**
- **SubjectCodeGroup nameAssessmentTitle.mp4**

References

Zhao, J., Gui, X. & Zhang, X. (2018). Deep convolution neural networks for Twitter sentiment analysis. *IEEE Access*, 6, 23253–23260. Retrieved from <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8244338>

Academic Integrity Declaration

We declare that except where we have referenced, the work we are submitting for this assessment task is our own work. We have read and are aware of Torrens University Australia Academic Integrity Policy and Procedure viewable online at <http://www.torrens.edu.au/policies-and-forms>

We are aware that we need to keep a copy of all submitted material and their drafts, and we will do so accordingly.

Assessment Rubric

Assessment Attributes	Fail (Yet to achieve minimum standard) 0–49%	Pass (Functional) 50–64%	Credit (Proficient) 65– 74%	Distinction (Advanced) 75– 84%	High Distinction (Exceptional) 85– 100%
<i>Problem Statement and Aim</i> - Context of study - Problem statement - Specific aims or questions of study - Link to relevant body of knowledge - Practical relevance and usefulness Percentage for this criterion = 20%	Context and problem statement is not properly defined. Research aims or questions are unclear and/or are not properly justified. No clear links to the relevant body of knowledge.	Context, problem statement and aims or questions are defined but are loosely coupled. The above have high level implicit links to the relevant body of knowledge.	Context and problem statement is clearly defined, but the proposed aim and question are not specific to the defined problem. Justification is provided and attempts have been made to link the proposed questions to the relevant body of knowledge.	Problem, context, aim and research question(s) are clearly stated. The proposed aim and questions are well justified. The proposed aim and questions are somewhat linked to the relevant body of knowledge.	The problem statement clearly captures the nuances of the problem succinctly and sets up the proposal in a professional manner. The proposed aim and questions are clearly linked to the relevant body of knowledge.
<i>Analysis, Synthesis and Application of Literature</i> - Analysis and synthesis of existing knowledge and related theory - Application of existing literature - Project identification Percentage for this criterion = 25%	The analysis shows poor paraphrasing skills (content is mostly plagiarised). The synthesis of ideas is weak and inconsistent. The project is not clearly identified.	The analysis shows inconsistencies. The synthesis of ideas is inconsistent. The project is identified but is not clearly explained.	The review and the appraisal of the relevant literature is limited. The synthesis of ideas is fair. The project is identified but is not clearly explained.	The review appraises the relevant literature. The synthesis of ideas is good. The project is identified and is clearly explained.	The review critically appraises the relevant literature. The synthesis of ideas is comprehensive. The project is identified and is clearly explained.

<i>Project Planning and Management</i> • <i>Plan of activities</i> • <i>Risk analysis</i> • <i>Contingency Plan</i> • <i>Measures to ensure success of project</i> Percentage for this criterion = 25%	The proposed plan is not feasible. The plan does not account for the project constraints, time frame, procurement and/or budget. No risk analysis or contingency plan provided.	The proposed plan is feasible but lacks convincing statements. The plan does account for the project constraints, timeframe, procurement and/or budget at a very high level. The risk analysis and contingency plan provided is poor.	Presents a feasible plan, including a basic description of the constraints, time frame and budget. Some attempt to analyse the risks, set up a contingency plan and identify critical success factors in relation to the project.	Presents a detailed research plan that is clearly feasible within the time frame, budget and other constraints. Sound analysis of the risks associated with the project. Sound contingency plan and critical success factors proposed.	Shows good use of project management processes to plan and manage the project work. Undertakes an excellent review of the risks, contingency plan and critical success factors in relation to the project.
<i>Overall Structure and Flow of Proposal</i> • <i>Required sections (abstract, introduction, body and conclusion)</i> • <i>Organisation and structure</i> • <i>Appropriate argument</i> Percentage for this criterion = 15%	Writing is difficult to read or follow. Ideas do not link up well within and/or between paragraphs. A flawed argument is presented. No acknowledgement of the audience.	Writing is not consistent, as ideas are only loosely linked within and/or between paragraphs. A mediocre argument is presented. The acknowledgement of the audience is too wide.	Writing is easy to follow and clear in meaning. Only minor spelling or grammar mistakes. Some effort made to tailor the argument and the evidence for the audience.	The report is well organised and the ideas and arguments flow naturally and logically. The argument and the evidence are appropriate for the target audience.	The report builds a strong and compelling argument for the next assessment and further research and is targeted at the appropriate audience.

Audio-visual presentation • <i>Delivery</i> • <i>Content organisation</i> • <i>Audience awareness</i> Percentage for this criterion = 15%	Speaks in a low volume and/or monotonous tone, which causes audience to disengage.	Speaks in an uneven volume with little or no inflection.	Speaks in an uneven volume with little inflection.	Speaks with satisfactory variation in volume and inflection.	Speaks with fluctuation in volume and inflection to maintain audience interest and emphasize key points.
	Does not clearly define subject and purpose; provides weak or no support of subject; gives insufficient support for ideas or conclusions. Fails to increase audience understanding or their knowledge of the topic.	Attempts to define the purpose and the subject; provides irrelevant examples, facts and/or statistics, which do not adequately support the subject; includes very thin data or evidence Increases audience understanding and knowledge of at least one point.	Attempts to define the purpose and the subject; provides weak examples, facts and/or statistics, which do not adequately support the subject; includes very thin data or evidence. Increases audience understanding and knowledge of some points.	Purpose and subject are somewhat clear; includes some examples, facts and/or statistics that support the subject; includes some data or evidence that supports the conclusions. Increases audience understanding and awareness of most points.	The purpose and subject are clear; includes pertinent examples, facts and/or statistics; supports conclusions/ideas with evidence. Significantly increases audience understanding and knowledge of the topic; convinces the audience to recognise the validity and importance of the subject.

The following Subject Learning Outcomes are addressed in this assessment	
SLO b)	Compare and select ways to pre-process signals, images, and texts for natural language, speech recognition, and computer vision applications.
SLO c)	Develop critical analysis skills in deep learning research.
SLO d)	Demonstrate collaborative skills to apply deep learning to solve real world problems.
SLO e)	Demonstrate ability to effectively communicate scientific and technical information.

